

54L Series — Metal Oxide Varistors

54 mm class MOVs for high-energy DC surge positions · Datasheet · Rev. 2026-10-21

Description

The 54L series is built on the largest varistor discs in the Flyboat range (54 mm class), for high-energy surge positions in DC power systems — photovoltaic plants, DC distribution and energy storage. Unlike the other series, 54L is not published with a full catalog ratings table; data for individual grades is available on request. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.

Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- 54 mm class — the largest varistor discs in the Flyboat range
- One published DC model: 54LK911, varistor voltage 910 V
- 54LK911 is UL 1449 Recognized for DC use, File E335803
- Built for high-energy surge positions in DC power systems
- Individual grade data available on request
- Discs pressed, sintered, metallized and tested in our own plant

Agency approvals

One model is published for this series: 54LK911 (varistor voltage 910 V), UL 1449 Recognized for DC use, File E335803. Verify in UL Product iQ (productiq.ulprospector.com) by searching file number E335803 — the record lists exactly which model codes are covered.

Model naming and standard marking

Model naming: 54L K 431 — 54 = nominal size (mm), L = rectangular; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default). Standard marking: the manufacturer logo and the model code are etched on the encapsulation.

Published grade

Model	Varistor voltage	Recognition
54LK911	910 V	UL 1449 Recognized (DC), File E335803

Full electrical data for this grade — maximum continuous DC voltage, clamping voltage, discharge and energy ratings — is available on request.

Ordering and support: samples and design-in support — North America: Thomas Dong, Flyboat Electronic LLC, Bellevue, WA (thomasd@flyboatmov.com); China: Colin Fei (colinf@flyboatmov.com). Specifications are subject to change without notice; verify critical ratings against the model row above before design freeze.